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Network device and method with bypass capability

Abstract

A system for a building includes a device includes a first and second connector. The device also includes a switch having a first, a second, and a third port. The first connector is coupled to the first port of the switch and the second connector is coupled to the second port of the switch. The device also includes a controller that causes the first port to be coupled to the second port when the switch is not involved in communication between the first and second ports and the third port. When the switch is involved in communication between the first and second ports, the first port and the second port are each coupled to the third port, and the third port is coupled to the controller.

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Background/Summary

BACKGROUND

(1) The present disclosure relates generally to networks. The present disclosure relates more particularly to a bypass system or method for a network, such as an Ethernet network.

(2) A building management system (BMS) is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS can include a heating, ventilation, or air conditioning (HVAC) system, a security system, a lighting system, a fire alerting system, another system that is capable of managing building functions or devices, or any combination thereof. BMS devices may be installed in any environment (e.g., an indoor area or an outdoor area) and the environment may include any number of buildings, spaces, zones, rooms, or areas. A BMS may include METASYS® or VERASYS® building controllers or other devices sold by Johnson Controls, Inc., as well as building devices and components from other sources.

(3) A BMS often utilizes a network (e.g., an Ethernet network, a Lon works Network, a building automation and control network (BACnet)) for communications. Networks often communicate by passing messages through one or more intermediate devices between the source device and the destination device. If an intermediate device fails or is powered off, the communication chain can be broken.

SUMMARY OF THE INVENTION

- (4) One embodiment of the present disclosure relates to a building system of a building. The building system includes a building controller configured to operate one or more pieces of building equipment of the building. The building system further includes controllers connected to Ethernet, where the controllers are configured in a daisy chain with bypass capability to prevent a loss of function in the event of a power failure.
- (5) In some embodiments, the building controller and the IP controller are separate units connected via one or more wires. In some embodiments, the controller contains Ethernet jacks, transformers, an Ethernet switch, and a microcontroller.
- (6) In some embodiments, the transformers are controlled by the microcontroller. In some other embodiments, magnetic circuits are placed between the Ethernet jacks and the microcontroller. The Ethernet jacks are coupled to the magnetic circuits.
- (7) In some embodiments, the magnetic circuits contain two circuits containing two transformers each. In other embodiments, the magnetic circuits contain four circuits containing two transformers each. In some embodiments, each circuit contains a second transformer that uses a half core. In some embodiments, the half core is made of a ferrous material. In some embodiments, a magnetic circuit uses one or more windings around a ferrous core, and the ferrous core switchable between one or more of the windings to achieve a by bypass operation.
- (8) In some embodiments, the half core can switch between communicating with a first wire coil and a second wire coil. The switch in communication is guided by whether a switching coil is in an on or off position. In some embodiments, when the switching coil is on, the magnetics behave with normal function and the half core interacts with the first wire coil. In some embodiments, the first wire coil is coupled to the Ethernet switch and microcontroller.
- (9) In other embodiments, when the switching coil is off, the half core interacts with the second wire coil. In some embodiments, the second wire coil is coupled to a wire coil in a second magnetics component that is connected to a second Ethernet jack.
- (10) According to various embodiments, a system for a building includes a device that includes a first and second connector. The device also includes a switch having a first, a second, and a third port. The first connector is coupled to the first port of the switch and the second connector is coupled to the second port of the switch. The device also includes a controller that causes the first port to be coupled to the second port when the switch is not involved in communication between the first and second ports and the third port. When the switch is involved in communication between the first and second ports, the first port and the second port are each coupled to the third port, and the third port is coupled to the controller.
- (11) In some embodiments, the switch includes a number of transformer circuits. Each circuit includes a primary coil that is coupled to the first connector or second connector, a first secondary coil that is coupled to the third port, and a second secondary coil that is coupled to a third secondary coil of another transformer circuit.
- (12) In some embodiments, the switch also includes a movable core that interacts with the first secondary coil when the switch is involved in communication between the first and second ports.
- (13) In some embodiments, the movable core interacts with the second secondary coil when the switch is not involved in communication between the first and second ports. In some embodiments, the movable core is moved by the controller. In some embodiments, the controller contains a plurality of magnetic circuits. In some embodiments, the switch comprises four transformers. In some embodiments, the first and second connectors are Ethernet jacks.
- (14) According to various embodiments, a method of bypassing a communication controller in a network of devices includes moving a core of a transformer to a first position when a first device is operational. When the core is in the first position, signals are communicated from a first connector through the first device to a second connector. The method also includes moving the core of the transformer to a second position when the first device is not operational. When the core is in the

second position, the signals are communicated from the first connector to the second connector without reaching the first device.

(15) In some embodiments, the core is biased to be in the second position and is moved to the first position by magnetism. In some embodiments, the transformer is enclosed in an Ethernet connector. In some embodiments, the transformer includes a first pair of transmit output conductors and two pairs of transmit input conductors. In some embodiments, the transformer includes a first pair of receive input conductors and two pairs of receive output conductors.

(16) According to various embodiments, a communication switch for a network includes a first device and a second device. The communication switch includes a first magnetic circuit that includes a first transmit output that is configured to be coupled to a first connector, a first transmit input, and a second transmit input. The switch also includes a second magnetic circuit that includes a first receive input that is configured to be coupled to the first connector, a first receive output, and a second receive output. The switch also includes a third magnetic circuit that includes a second transmit output that is configured to be coupled to a second connector, a third transmit input, and a fourth transmit input, wherein the fourth transmit input is configured to be coupled to the second transmit input, the first transmit input and the third transmit input configured to be coupled to a controller. The switch also includes a fourth magnetic circuit that includes a second receive input that is configured to be coupled to the second connector, a third receive output, and a fourth receive output, wherein the fourth receive output is configured to be coupled to the second receive output, the first receive output and the third receive output configured to be coupled to the controller. In some embodiments, the first transmit output is configured to be coupled to the first transmit input and the first receive output is configured to be coupled to the first receive input in a first condition. The first transmit output is configured to be coupled to the second transmit input and the first receive input is configured to be coupled to the second receive output in a second condition.

(17) In some embodiments, the second condition is a power loss in the controller. In some embodiments, the first transmit output and the first receive input are coupled to a first Ethernet jack. In some embodiments, the second transmit output and the second receive input are coupled to a second Ethernet jack. In some embodiments, the first, second, third, and fourth magnetic circuits each include a movable core. In some embodiments, the first, second, third, and fourth magnetic circuits each include a transformer with a movable core. In some embodiments, the communications switch is a three port Ethernet switch.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

(2) FIG. 1 is a schematic drawing of a building equipped with a heating, ventilating, and air conditioning (HVAC) system, according to an exemplary embodiment.

(3) FIG. 2 is a block diagram of a building management system (BMS) which can be used to monitor and control the building and HVAC system of FIG. 1, according to an exemplary embodiment.

(4) FIG. 3 is a block diagram illustrating a system manager, zone coordinator, and zone controller of the BMS of FIG. 2 in greater detail, according to an exemplary embodiment.

(5) FIG. 4 is a block diagram of a network of controllers in communication with each other for use in the systems of FIG. 1 or 2.

(6) FIG. 5 is a more detailed block diagram of one of the controllers of FIG. 4.

(7) FIG. 6 is an electrical schematic drawing of the magnetic circuit of the controller of FIG. 5.

(8) FIG. 7 is an electrical schematic drawing of the magnetic circuit of the controller of FIG. 5.

(9) FIG. 8 is an electrical schematic drawing of the magnetic circuit of the controller of FIG. 5.

DETAILED DESCRIPTION

(10) Overview

(11) Referring generally to the FIGURES, a network (e.g., a network connected by a daisy chain, an Ethernet daisy chain) includes building controllers (e.g., building equipment such as air handling units, roof top units, zone controllers, thermostats, smart dampers, etc.) according to exemplary embodiments. Building controllers (e.g., configured as IP controllers) may be wired with Ethernet installation. By configuring Ethernet controllers in a daisy chain to have a bypass circuit, power failures do not shut down the function and communications other controllers on the network in some embodiments. For example, if one controller experiences a power failure, communications bypass that controller and the communications use the next controller that is functioning normally.

(12) In some embodiments, the controllers include a switch unit that includes a magnetic circuit for providing bypass operations. The magnetic circuit includes a transformer with a movable coil in some embodiments. The switch unit includes two ports associated with connectors to the network and one port coupled to a microcontroller or communications control circuit of the controller. The bypass circuit connects the two ports when the controller is inoperable (e.g., due to power loss).

(13) Building and HVAC System

(14) Referring now to FIG. 1, an exemplary building and HVAC system in which the systems and methods of the present invention can be implemented are shown, according to an exemplary embodiment. In FIG. 1, a perspective view of a building **10** is shown. Building **10** is served by a HVAC system **100**. HVAC system **100** can include HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage, etc.) configured to provide heating, cooling, ventilation, or other services for building **10**. The HVAC devices can be coupled together via a network as described below. For example, HVAC system **100** is shown to include a waterside system **120** and an airside system **130**. Waterside system **120** can provide a heated or chilled fluid to an air handling unit of airside system **130**. Airside system **130** can use the heated or chilled fluid to heat or cool an airflow provided to building **10**.

(15) HVAC system **100** is shown to include a chiller **102**, a boiler **104**, and a rooftop air handling unit (AHU) **106**. Waterside system **120** can use boiler **104** and chiller **102** to heat or cool a working fluid (e.g., water, glycol, etc.) and can circulate the working fluid to AHU **106**. In various embodiments, the HVAC devices of waterside system **120** can be located in or around building **10** (as shown in FIG. 1) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant, etc.). The working fluid can be heated in boiler **104** or cooled in chiller **102**, depending on whether heating or cooling is required in building **10**. Boiler **104** can add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. Chiller **102** can place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from chiller **102** and/or boiler **104** can be transported to AHU **106** via piping **108**.

(16) AHU **106** can place the working fluid in a heat exchange relationship with an airflow passing through AHU **106** (e.g., via one or more stages of cooling coils and/or heating coils). The airflow can be, for example, outside air, return air from within building **10**, or a combination of both. AHU **106** can transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow. For example, AHU **106** can include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid can then return to chiller **102** or boiler **104** via piping **110**.

(17) Airside system **130** can deliver the airflow supplied by AHU **106** (i.e., the supply airflow) to building **10** via air supply ducts **112** and can provide return air from building **10** to AHU **106** via air return ducts **114**. In some embodiments, airside system **130** includes multiple variable air volume (VAV) units **116**. For example, airside system **130** is shown to include a separate VAV unit **116** on each floor or zone of building **10**. VAV units **116** can include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of building **10**. In other embodiments, airside system **130** delivers the supply airflow into one or more zones of building **10** (e.g., via supply ducts **112**) without using intermediate VAV units **116** or other flow control elements. AHU **106** can include various sensors (e.g., temperature sensors, pressure sensors, etc.) configured to measure attributes of the supply airflow. AHU **106** can receive input from sensors located within AHU **106** and/or within the building zone and can adjust the flow rate, temperature, or other attributes of the supply airflow through AHU **106** to achieve setpoint conditions for the building zone.

(18) Building Management System

(19) Referring now to FIG. 2, a block diagram of a building management system (BMS) **200** is shown, according to an exemplary embodiment. A BMS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS **200** can include, for example, a HVAC system, a security system, a lighting system, and a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof. BMS **200** can be used to monitor and control the devices of HVAC system **100** and/or airside system (e.g., HVAC equipment) as well as other types of BMS devices (e.g., lighting equipment, security equipment, etc.).

(20) In brief overview, BMS **200** provides a system architecture that facilitates automatic equipment discovery and equipment model distribution. Equipment discovery can occur on multiple levels of BMS **200** across multiple different communications busses (e.g., a system bus **254**, zone buses **256-260** and **264**, sensor/actuator bus **266**, etc.) and across multiple different communications protocols. In some embodiments, equipment discovery is accomplished using active node tables, which provide status information for devices connected to each communications bus. For example, each communications bus can be monitored for new devices by monitoring the corresponding active node table for new nodes. When a new device is detected, BMS **200** can begin interacting with the new device (e.g., sending control signals, using data from the device) without user interaction.

(21) Some devices in BMS **200** present themselves to the network using equipment models. An equipment model defines equipment object attributes, view definitions, schedules, trends, and the associated BACnet value objects (e.g., analog value, binary value, multistate value, etc.) that are used for integration with other systems. An equipment model for a device can include a collection of point objects that provide information about the device (e.g., device name, network address, model number, device type, etc.) and store present values of variables or parameters used by the device. For example, the equipment model can include point objects (e.g., standard BACnet point objects) that store the values of input variables accepted by the device (e.g., setpoint, control parameters, etc.), output variables provided by the device (e.g., temperature measurement, feedback signal, etc.), configuration parameters used by the device (e.g., operating mode, actuator stroke length, damper position, tuning parameters, etc.). The point objects in the equipment model can be mapped to variables or parameters stored within the device to expose those variables or parameters to external systems or devices.

(22) Some devices in BMS **200** store their own equipment models. Other devices in BMS **200** have equipment models stored externally (e.g., within other devices). For example, a zone coordinator **208** can store the equipment model for a bypass damper **228**. In some embodiments, zone coordinator **208** automatically creates the equipment model for bypass damper **228** or other devices on zone bus **258**. Other zone coordinators can also create equipment models for devices connected

to their zone busses. The equipment model for a device can be created automatically based on the types of data points exposed by the device on the zone bus, device type, and/or other device attributes. Several examples of automatic equipment discovery and equipment model distribution are discussed in greater detail below.

(23) Still referring to FIG. 2, BMS **200** is shown to include a system manager **202**; several zone coordinators **206**, **208**, **210** and **218**; and several zone controllers **224**, **230**, **232**, **236**, **248**, and **250**. System manager **202** can communicate with client devices **204** (e.g., user devices, desktop computers, laptop computers, mobile devices, etc.) via a data communications link **374** (e.g., BACnet IP, Ethernet, wired or wireless communications, etc.). System manager **202** can provide a user interface to client devices **204** via data communications link **374**. The user interface may allow users to monitor and/or control BMS **200** via client devices **204**.

(24) In some embodiments, system manager **202** is connected with zone coordinators **206-210** and **218** via a system bus **254**. System bus **254** can include any of a variety of communications hardware (e.g., wire, optical fiber, terminals, etc.) configured to facilitate communications between system manager and other devices connected to system bus **254**. Throughout this disclosure, the devices connected to system bus **254** are referred to as system bus devices. System manager **202** can be configured to communicate with zone coordinators **206-210** and **218** via system bus **254** using a master-slave token passing (MSTP) protocol or any other communications protocol. System bus **254** can also connect system manager **202** with other devices such as a constant volume (CV) rooftop unit (RTU) **212**, an input/output module (IOM) **214**, a thermostat controller **216** (e.g., a TEC2000 series thermostat controller), and a network automation engine (NAE) or third-party controller **220**. RTU **212** can be configured to communicate directly with system manager **202** and can be connected directly to system bus **254**. Other RTUs can communicate with system manager **202** via an intermediate device. For example, a wired input **262** can connect a third-party RTU **242** to thermostat controller **216**, which connects to system bus **254**.

(25) System manager **202** can provide a user interface for any device containing an equipment model. Devices such as zone coordinators **206-210** and **218** and thermostat controller **216** can provide their equipment models to system manager **202** via system bus **254**. In some embodiments, system manager **202** automatically creates equipment models for connected devices that do not contain an equipment model (e.g., IOM **214**, third party controller **220**, etc.). For example, system manager **202** can create an equipment model for any device that responds to a device tree request. The equipment models created by system manager **202** can be stored within system manager **202**. System manager **202** can then provide a user interface for devices that do not contain their own equipment models using the equipment models created by system manager **202**. In some embodiments, system manager **202** stores a view definition for each type of equipment connected via system bus **254** and uses the stored view definition to generate a user interface for the equipment.

(26) Each zone coordinator **206-210** and **218** can be connected with one or more of zone controllers **224**, **230-232**, **236**, and **248-250** via zone busses **256**, **258**, **260**, and **264**. Zone busses **256**, **258**, **260**, and **264** can include any of a variety of communications hardware (e.g., wire, optical fiber, terminals, etc.) configured to facilitate communications between a zone coordinator and other devices connected to the corresponding zone bus. Throughout this disclosure, the devices connected to zone busses **256**, **258**, **260**, and **264** are referred to as zone bus devices. Zone coordinators **206-210** and **218** can communicate with zone controllers **224**, **230-232**, **236**, and **248-250** via zone busses **256-260** and **264** using a MSTP protocol or any other communications protocol. Zone busses **256-260** and **264** can also connect zone coordinators **206-210** and **218** with other types of devices such as variable air volume (VAV) RTUs **222** and **240**, changeover bypass (COBP) RTUs **226** and **252**, bypass dampers **228** and **246**, and PEAK controllers **234** and **244**.

(27) Zone coordinators **206-210** and **218** can be configured to monitor and command various zoning systems. In some embodiments, each zone coordinator **206-210** and **218** monitors and

commands a separate zoning system and is connected to the zoning system via a separate zone bus. For example, zone coordinator **206** can be connected to VAV RTU **222** and zone controller **224** via zone bus **256**. Zone coordinator **208** can be connected to COBP RTU **226**, bypass damper **228**, COBP zone controller **230**, and VAV zone controller **232** via zone bus **258**. Zone coordinator **210** can be connected to PEAK controller **234** and VAV zone controller **236** via zone bus **260**. Zone coordinator **218** can be connected to PEAK controller **244**, bypass damper **246**, COBP zone controller **248**, and VAV zone controller **250** via zone bus **264**.

(28) A single model of zone coordinator **206-210** and **218** can be configured to handle multiple different types of zoning systems (e.g., a VAV zoning system, a COBP zoning system, etc.). Each zoning system can include a RTU, one or more zone controllers, and/or a bypass damper. For example, zone coordinators **206** and **210** are shown as Verasys VAV engines (VVEs) connected to VAV RTUs **222** and **240**, respectively. Zone coordinator **206** is connected directly to VAV RTU **222** via zone bus **256**, whereas zone coordinator **210** is connected to a third-party VAV RTU **240** via a wired input **268** provided to PEAK controller **234**. Zone coordinators **208** and **218** are shown as Verasys COBP engines (VCEs) connected to COBP RTUs **226** and **252**, respectively. Zone coordinator **208** is connected directly to COBP RTU **226** via zone bus **258**, whereas zone coordinator **218** is connected to a third-party COBP RTU **252** via a wired input **270** provided to PEAK controller **244**.

(29) Zone controllers **224**, **230-232**, **236**, and **248-250** can communicate with individual BMS devices (e.g., sensors, actuators, etc.) via sensor/actuator (SA) busses. For example, VAV zone controller **236** is shown connected to networked sensors **238** via SA bus **266**. Networked sensors **238** can include, for example, temperature sensors, humidity sensors, pressure sensors, lighting sensors, security sensors, or any other type of device configured to measure and/or provide an input to zone controller **236**. Zone controller **236** can communicate with networked sensors **238** using a MSTP protocol or any other communications protocol. Although only one SA bus **266** is shown in FIG. 2, it should be understood that each zone controller **224**, **230-232**, **236**, and **248-250** can be connected to a different SA bus. Each SA bus can connect a zone controller with various sensors (e.g., temperature sensors, humidity sensors, pressure sensors, light sensors, occupancy sensors, etc.), actuators (e.g., damper actuators, valve actuators, etc.) and/or other types of controllable equipment (e.g., chillers, heaters, fans, pumps, etc.).

(30) Each zone controller **224**, **230-232**, **236**, and **248-250** can be configured to monitor and control a different building zone. Zone controllers **224**, **230-232**, **236**, and **248-250** can use the inputs and outputs provided via their SA busses to monitor and control various building zones. For example, a zone controller **236** can use a temperature input received from networked sensors **238** via SA bus **266** (e.g., a measured temperature of a building zone) as feedback in a temperature control algorithm. Zone controllers **224**, **230-232**, **236**, and **248-250** can use various types of control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control a variable state or condition (e.g., temperature, humidity, airflow, lighting, etc.) in or around building **10**.

(31) Referring now to FIG. 3, a block diagram illustrating a portion of BMS **200** in greater detail is shown, according to an exemplary embodiment. BMS **200** is shown to include system manager **202**, a zone coordinator **308**, and a zone controller **322**. Zone coordinator **308** can be any of zone coordinators **206-210** or **218**. Zone controller **322** can be any of zone controllers **224**, **230**, **232**, **236**, **248**, or **250**. Zone coordinator **308** can be connected with system manager via system bus **254**. For example, system bus **254** is shown connecting a first system bus datalink **304** within system manager **202** with a second system bus datalink **310** within zone coordinator **308**. Zone coordinator **308** can be connected with zone controller **322** via a zone bus **318**. For example, zone bus **318** is shown connecting a first zone bus datalink **314** within zone coordinator **308** with a second zone bus

datalink **320** within zone controller **322**. Zone bus **318** can be any of zone busses **256-260** or **264**. Zone controller **322** is connected with networked sensors **238** and actuators **332** via a SA bus **266**. (32) BMS **200** can automatically discover new equipment connected to any of system bus **254**, zone bus **318**, and SA bus **266**. Advantageously, the equipment discovery can occur automatically (e.g., without user action) without requiring the equipment to be placed in discovery mode and without sending a discovery command to the equipment. In some embodiments, the automatic equipment discovery is based on active node tables for system bus **254**, zone bus **318**, and SA bus **266**. Each active node table can provide status information for the devices communicating on a particular bus. For example, the active node table **306** for system bus **254** can indicate which MSTP devices are participating in the token ring used to exchange information via system bus **254**. Active node table **306** can identify the devices communicating on system bus **254** by MAC address or other device identifier. Devices that do not participate in the token ring (e.g., MSTP slave devices) can be automatically discovered using a net sensor plug and play (described in greater detail below).

(33) The active node table for each communications bus can be stored within one or more devices connected to the bus. For example, active node table **306** can be stored within system manager **202**. In some embodiments, active node table **306** is part of a system bus datalink **304** (e.g., a MSTP datalink) used by system manager **202** to communicate via system bus **254**. System manager **202** can subscribe to changes in value of active node table **306** and can receive a notification (e.g., from system bus datalink **304**) when a change in active node table **306**. In response to a notification that a change in active node table **306** has occurred, system manager **202** can read active node table **306** to detect and identify the devices connected to system bus **254**.

(34) In some embodiments, a device list generator **302** within system manager **202** generates a list of the devices connected to system bus **254** (i.e., a device list) based on active node table **306** and stores the device list within system manager **202**. The device list generated by system manager **202** can include information about each device connected to system bus **254** (e.g., device type, device model, device ID, MAC address, device attributes, etc.). When a new device is detected on system bus **254**, system manager **202** can automatically retrieve the equipment model from the device if the device stores its own equipment model. If the device does not store its own equipment model, system manager **202** can retrieve a list of point values provided by the device. System manager **202** can then use the equipment model and/or list of point values to present information about the connected system bus devices to a user.

(35) The active node tables for each zone bus can be stored within the zone coordinator connected to the zone bus. For example, the active node table **316** for zone bus **318** can be stored within zone coordinator **308**. In some embodiments, active node table **316** is part of a zone bus datalink **314** (e.g., a MSTP datalink) used by the zone coordinator **308** to communicate via zone bus **318**. Zone coordinator **308** can subscribe to changes in value of active node table **316** and can receive a notification (e.g., from zone bus datalink **314**) when a change in active node table **316** occurs. In response to a notification that a change to active node table **316** has occurred, zone coordinator **308** can read active node table **316** to identify the devices connected to zone bus **318**.

(36) In some embodiments, a detector object **312** of zone coordinator **308** generates a list of the devices communicating on zone bus **318** (i.e., a device list) based on active node table **316** and stores the device list within zone coordinator **308**. Each zone coordinator in BMS **200** can generate a list of devices on the connected zone bus. The device list generated by each zone coordinator **308** can include information about each device connected to zone bus **318** (e.g., device type, device model, device ID, MAC address, device attributes, etc.). When a new device is detected on zone bus **318**, the connected zone coordinator **308** can automatically retrieve the equipment model from the device if the device stores its own equipment model. If the device does not store its own equipment model, the connected zone coordinator **308** can retrieve a list of point values provided by the device.

(37) Zone coordinator **308** can incorporate the new zone bus device into the zoning logic and can inform system manager **202** that a new zone bus device has been added. For example, zone coordinator **308** is shown providing a field device list to system manager **202**. The field device list can include a list of devices connected to zone bus **318** and/or SA bus **266**. System manager **202** can use the field device list and the list of system bus devices to generate a device tree including all of the devices in BMS **200**. In some embodiments, zone coordinator **308** provides an equipment model for a connected zone bus device to system manager **202**. System manager **202** can then use the equipment model and/or list of point values for the new zone bus device to present information about the new zone bus device to a user.

(38) In some embodiments, the device list generated by each zone coordinator **308** indicates whether system manager **202** should communicate directly with the listed zone bus device (e.g., VAV RTU **222**, VAV zone controller **224**, etc.) or whether system manager **202** should communicate with the intermediate zone coordinator **308** on behalf of the zone bus device. In some embodiments, system manager **202** communicates directly with zone bus devices that provide their own equipment models, but communicates with the intermediate zone coordinator **308** for zone bus devices that do not provide their own equipment model. As discussed above, the equipment models for zone bus devices that do not provide their own equipment model can be generated by the connected zone coordinator **308** and stored within the zone coordinator **308**. Accordingly, system manager **202** may communicate directly with the device that stores the equipment model for a connected zone bus device (i.e., the zone bus device itself or the connected zone coordinator **308**).

(39) The active node table **330** for SA bus **266** can be stored within zone controller **322**. In some embodiments, active node table **330** is part of the SA bus datalink **328** (e.g., a MSTP datalink) used by zone controller **322** to communicate via SA bus **266**. Zone controller **322** can subscribe to changes in value of the active node table **330** and can receive a notification (e.g., from SA bus datalink **328**) when a change in active node table **330** occurs. In response to a notification that a change to active node table **330** has occurred, zone controller **322** can read active node table **330** to identify some or all of the devices connected to SA bus **266**. In some embodiments, active node table **330** identifies only the SA bus devices participating in the token passing ring via SA bus **266** (e.g., MSTP master devices). Zone controller **322** can include an additional net sensor plug and play (NsPnP) **324** configured to detect SA bus devices that do not participate in the token passing ring (e.g., MSTP slave devices).

(40) In some embodiments, NsPnP **324** is configured to actively search for devices connected to SA bus **266** (e.g., networked sensors **238**, actuators **332**, lighting controllers **334**, etc.). For example, NsPnP **324** can send a “ping” to a preconfigured list of MSTP slave MAC addresses. For each SA bus device that is discovered (i.e. responds to the ping), NsPnP **324** can dynamically bring it online. NsPnP **324** can bring a device online by creating and storing an instance of a SA bus device object representing the discovered SA bus device. NsPnP **324** can automatically populate the SA bus device object with all child point objects needed to collect and store point data (e.g., sensor data) from the newly-discovered SA bus device. In some embodiments, NsPnP **324** automatically maps the child point objects of the SA bus device object to attributes of the equipment model for zone controller **322**. Accordingly, the data points provided by the SA bus devices can be exposed to zone coordinator **308** and other devices in BMS **200** as attributes of the equipment model for zone controller **322**.

(41) In some embodiments, a detector object **326** of zone controller **322** generates a list of the devices connected to SA bus **266** (i.e., a device list) based on active node table **330** and stores the device list within zone controller **322**. NsPnP **324** can update the device list to include any SA bus devices discovered by NsPnP **324**. The device list generated by zone controller **322** can include information about each device connected to SA bus **266** (e.g., device type, device model, device ID, MAC address, device attributes, etc.). When a new device is detected on SA bus **266**, zone controller **322** can automatically retrieve the equipment model from the device if the device stores

its own equipment model. If the device does not store its own equipment model, zone controller **322** can retrieve a list of point values provided by the device.

(42) Zone controller **322** can incorporate the new SA bus device into the zone control logic and can inform zone coordinator **308** that a new SA bus device has been added. Zone coordinator **308** can then inform system manager **202** that a new SA bus device has been added. For example, zone controller **322** is shown providing a SA device list to zone coordinator **308**. The SA device list can include a list of devices connected to SA bus **266**. Zone coordinator **308** can use the SA device list and the detected zone bus devices to generate the field device list provided to system manager **202**. In some embodiments, zone controller **322** provides an equipment model for a connected SA bus device to zone coordinator **308**, which can be forwarded to system manager **202**. System manager **202** can then use the equipment model and/or list of point values for the new SA bus device to present information about the new SA bus device to a user. In some embodiments, data points provided by the SA bus device are shown as attributes of the zone controller **322** to which the SA bus device is connected.

(43) Additional features and advantages of BMS **200**, system manager **202**, zone coordinator **308**, and zone controller **322** are described in detail in U.S. patent application Ser. No. 15/179,894 filed Jun. 10, 2016, the entire disclosure of which is incorporated by reference herein.

(44) Network Connections

(45) Referring now to FIG. 4, a system **400** includes a number of controllers **450** (e.g., IP controllers), connectors **410**, **415** (Ethernet jacks) and wiring or cables **405** (Ethernet cables) according to an exemplary embodiment. The system **400** can be or be part of system **100** of the building **10** of FIG. 1 and/or implemented in the BMS **200** as described with reference to FIGS. 2-3. Controllers **450** can be any type of HVAC, security, lighting, safety, or network equipment that communicates via a wired network in some embodiments. In addition, controllers **450** can be non-HVAC equipment, such as any network device.

(46) The cables **405** link the individual controllers **450** in a daisy chain configuration in some embodiments. In some embodiments, each of controllers **450** includes connectors **410** and **415** where a connector **410** is coupled to an upstream controller and a connector **415** is coupled to a downstream controller. Controllers **450** are configured to provide a bypass which allows other controllers **450** to remain communicative in the event of a single or multi point controller power failure or other event that prevents a controller **450** from being communicative in some embodiments. The Ethernet system **400** may be part of data communications link **374** as described above.

(47) Referring now to FIG. 5, the controller **450** of FIG. 4 includes a switch unit **452**, a microcontroller **440** and connectors **410** and **415**. According to an exemplary embodiment, an end of a first cable **407** (similar to cable **405** (FIG. 4)) connects to controller **450** via connector **410**. Another cable **409** (similar to cable **405** (FIG. 4)) connects to connector **415** on the controller **450**. Connectors **410**, **415** are each connected to switch unit **452**.

(48) Switch unit **452** is a three port Ethernet-type switch unit in some embodiments. The first two ports are coupled to connectors **410** and **415**, respectively, and a third port is coupled to microcontroller **440**. Microcontroller **440** can be any type of processor, communication controller, microprocessor, and is part of controller **450** in some embodiments. Microcontroller **440** can control communication operations as well as HVAC operations in some embodiments. In some embodiments, switch unit **452** controls communication operations with or without contributions from microcontroller **440**.

(49) In some embodiments, switch unit **452** includes a magnetic circuit **420**, and a switch **430** (e.g., three port Ethernet switch). Magnetic circuit **420** is a two channel magnetic circuit within controller **450**. In some embodiments, the magnetic circuit **420** may be integrated on microcontroller **440**, disposed in between connectors **410**, **415** and microcontroller **440**, integrated with network switch **430**, integrated with connectors **410** and **415** or otherwise disposed to provide a bypass path. In

some embodiments, magnetic circuit **420** is configured to connect connectors **410** and **415** to each other when microcontroller **440** cannot participate in communications (e.g., due to a power outage) and is configured to connect connectors **410** and **415** to first and second ports **442** and **444**, respectively, of switch **430** and ultimately microcontroller **440** in some embodiments.

(50) Switch **430** and/or microcontroller **440** can include circuitry for processing, relaying, storing, transmitting and receiving signals on cable **405**. Physical layer operations, media access control functions and network processing can be performed by switch **430** and/or microcontroller **440** in some embodiments. Microcontroller **440** can also provide HVAC or other BMS functions for controller **450**.

(51) In some embodiments, magnetic circuit **420** is a switch circuit that uses solid state switches or micro relay switches to provide a bypass circuit instead of a magnetic components. In such an embodiment, the solid state switches or relay switches bypass switch **430** and connect connectors **410** and **415** in the event of a power failure.

(52) Referring now to FIG. **6**, the magnetic circuit **420** of controller **450** includes magnetics channels **500** and **600** and a coil control circuit **700**. Connector **410** on controller **450** is coupled to a first magnetics channel **500**. The connector **415** of controller **450** is coupled to a second magnetics channel **600**. In some embodiments, magnetic circuit **420** may be integrated on microcontroller **440**, disposed in between connectors **410**, **415** and microcontroller **440**, is integrated with network switch **430**, is integrated with connectors **410** and **415** or otherwise disposed to provide a bypass path. In some embodiments, the first magnetics channel **500** may be identical to or similar to the second magnetics channel **600**. In some embodiments, magnetics channels **500**, **600** of the same controller **450** lead to the same switch **430** and the same microcontroller **440**.

(53) According to the exemplary embodiment, the first magnetics channel **500** contains a first set **501** of transformer circuits. The first transformer circuit set **501** comprises: a first transformer **505** and a second transformer **510**. The second transformer **510** comprises: wire coils or windings **515**, **520**, and **525** and a ferrous half core **530** positioned between winding **515** and windings **520**, **525**. In some embodiments, windings **520** and **525** may be positioned proximate one another. Winding **515** may be considered to be a primary coil and windings **520**, **525** may be considered to be first and second secondary coils, respectively, according to some embodiments.

(54) The first magnetics channel **500** also contains a second set **502** of transformer circuits. The second transformer circuit set **502** comprises: a first transformer **550** and a second transformer **555**. The second transformer **555** comprises: wire coils or windings **560**, **565**, and **570** and a ferrous half core **575** positioned between winding **560** and windings **565**, **570**. In some embodiments, windings **565** and **570** may be positioned proximate one another. Winding **560** may be considered to be a primary coil and windings **565**, **570** may be considered to be first and second secondary coils, respectively, according to some embodiments. Windings **565** and **570** are coupled to port **442** in some embodiments.

(55) In some embodiments, the second magnetics channel **600** may be identical to the components described as being present in first magnetics channel **500**. The second magnetics channel **600** contains a first set **601** of transformer circuits. The first transformer circuit set **601** comprises: a first transformer **605** and a second transformer **610**. The second transformer **610** comprises: wire coils or windings **615**, **620**, and **625** and a ferrous half core **630** positioned between winding **615** and windings **620**, **625**. In some embodiments, windings **620** and **625** may be positioned proximate one another. Winding **615** may be considered to be a primary coil and windings **620**, **625** may be considered to be first and second secondary coils, respectively, according to some embodiments.

(56) The second magnetics channel **600** also contains a second set of transformer circuits **602**. The second transformer circuit set comprises: a first transformer **650** and a second transformer **655**. The second transformer **655** comprises: wire coils or windings **660**, **665**, and **670** and a ferrous half core **675** positioned between winding **660** and windings **665**, **670**. In some embodiments, windings **665**

and **670** may be positioned proximate one another. Winding **660** may be considered to be a primary coil and windings **665**, **670** may be considered to be first and second secondary coils, respectively, according to some embodiments. Windings **665** and **620** are coupled to port **444** in some embodiments. Transformers **505**, **550**, **605**, and **650** provide filtering and are optional.

(57) In some embodiments, magnetics channels **500**, **600** may contain a number of transformer circuits depending on the quantity of data and connections. For example, magnetics channels **500**, **600** may contain four sets of transformer circuits for a data quantity greater than one gigabit.

(58) In some embodiments, a first end **521** of winding **520** extends out of the first magnetics channel **500** toward switch **430** and/or microcontroller **440**. A second end **522** of winding **520** extends out of the first magnetics channel **500** toward switch **430** and/or microcontroller **440**.

(59) In some embodiments, a first end **526** of winding **525** extends out of the first magnetics channel **500** and is coupled to a first ending **626** of winding **625** of transformer **610**. A second end **527** of winding **525** extends out of the first magnetics channel **500** and is coupled to a second ending **627** of winding **625** of transformer **610**. In other embodiments, the first and second endings **526**, **527** of winding **525** may not be coupled to the first and second endings **626**, **627** of winding **625**. In other embodiments still, the first and second endings **526**, **527** of windings **525** may extend out of the first magnetics channel **500** toward 3 port Ethernet switch **430** and/or microcontroller **440**.

(60) In some embodiments, first end **566** of winding **565** extends out of the first magnetics channel **500** toward 3 port Ethernet switch **430** and/or microcontroller **440**. A second end **567** of winding **565** extends out of the first magnetics channel **500** toward 3 port Ethernet switch **430** and/or microcontroller **440**.

(61) In some embodiments, a first end **571** of winding **570** extends out of the first magnetics channel **500** and is coupled to a first ending **671** of winding **670** of transformer **610**. A second end **572** of winding **570** extends out of the first magnetics channel **500** and is coupled to a second ending **672** of winding **670** of transformer **610**. In other embodiments, the first and second endings **571**, **572** of winding **570** may not be coupled to the first and second endings **671**, **672** of winding **670**. In other embodiments still, the first and second endings **571**, **572** of windings **570** may extend out of the first magnetics channel **500** toward switch **430** and/or microcontroller **440**.

(62) In some embodiments, first end **621** of winding **620** extends out of the first magnetics channel **600** toward 3 port Ethernet switch **430** and/or microcontroller **440**. A second end **622** of winding **620** extends out of the first magnetics channel **600** toward switch **430** and/or microcontroller **440**.

(63) In some embodiments, first end **666** of winding **665** extends out of the first magnetics channel **600** toward switch **430** and/or microcontroller **440**. A second end **667** of winding **665** extends out of the first magnetics channel **600** toward switch **430** and/or microcontroller **440**.

(64) Magnetic circuit **420** also includes coil control circuit **700**. In some embodiments, coil control circuit **700** includes a switching coil **702** in an on or closed position. In other embodiments, switching coil **702** is in an off or open position. Further description of the operation of switching coil **702** is described herein. Coil control circuit **700** is controlled by signals from switch **430** or microcontroller **440**.

(65) Referring now to FIG. 7, a first configuration of the magnetic circuit **420** is shown. According to this exemplary embodiment, switching coil **702** is in an energized or non-energized state. According to the exemplary embodiment, coil control circuit **700** is supplying power to control switching coil **702** this embodiment, ferrous half cores **530**, **575**, **630**, **675** are in up positions **531**, **576**, **631**, **676**, respectively.

(66) According to the embodiment shown in FIG. 7, first ends **521**, **566**, **621**, **666** and second ends **522**, **567**, **622**, **667** of windings **520**, **565**, **620**, **665**, respectively, may all be active and extend out to switch **430** and/or microcontroller **440** and may deliver a signal to switch **430** and/or microcontroller **440**. FIG. 7 is an exemplary embodiment of the system **400** operating under normal conditions.

(67) Referring now to FIG. 8, a second configuration of the magnetic circuit 420 is shown. According to this exemplary embodiment, switching coil 702 is not energized. In this embodiment, ferrous half cores 530, 575, 630, 675 are in down position 532, 577, 631, and 677, respectively. Cores 539, 575, 639, and 675 are biased in the down position by gravity or a bias (e.g., a spring). In some embodiments, coil control circuit 700 is automatically depowered when microcontroller 440 loses power because the circuit 700 and microcontroller share the same power supply. In some embodiments, microcontroller 440 provides a control signal to control current through coil 702 such that coil is de-energized when microcontroller 440 or switch unit 452 (FIG. 4) cannot participate in communications.

(68) When the switching coil 702 is not energized, first windings 520, 565, 620, 665, first ends 521, 566, 621, 666, and second ends 522, 567, 622, 667 are no longer active or communicating with switch 430 and/or microprocessor or microcontroller 440. In some embodiments, this may be caused by a single or multi-point power failure.

(69) When ferrous half cores 530, 575, 630, and 675 are in the down positions 532, 577, 631, 677, respectively, second winding or coil 525 is linked to second winding 625 and second winding or coil 570 is linked to second winding 670. Second winding 525 is linked to second winding 625 via first ends 526, 626, and second ends 527, 627. Second winding 570 is linked to second winding 670 via first ends 571, 671 and second ends 572, 672. According to the exemplary embodiment, first end 526 may be considered to be the same as first ending 626, first end 571 may be considered to be the same as first ending 671, second ending 527 may be considered to be the same as second ending 627 and second ending 572 may be considered to be the same as second ending 672.

(70) In some embodiments, switching coil 702 is responsible for the movement of any of the ferrous half cores 530, 575, 630, 675 from the up position to the down position or vice versa. According to the exemplary embodiment, the presence of the two secondary coils of each second transformer 510, 555, 610, 655 causes a magnetic field created by each second transformer 510, 555, 610, 655 to move from the first secondary coil to the second secondary coil in each transformer 510, 555, 610, 655. Therefore, each second secondary coil 525, 570 626, 670 may be coupled to the ferrous half core in down positions 532, 577, 632, 677, respectively so that the magnetic circuit 420 of controller 450 may still function in the event of a power failure.

(71) According to the exemplary embodiment shown in FIG. 8, the systems or channels 500, 600 are coupled in such a way that a signal can be communicated and continue to traverse the daisy chain of controllers 450. For example, if a single controller 450 experiences a power failure, the coupling of secondary coils 525 to 626 and 570 to 670 creates a coupling of systems or channels 500 and 600, according to an exemplary embodiment. Furthermore, cables 405 plugged into connectors 410, 415 connect the controller 450 that experienced the power failure to two additional controllers 450 that have not experienced a power failure. Hence, the controller 450 that has experienced a power failure remains able to communicate necessary data to a switch 430, a microcontroller 440, or any other necessary components not described herein.

Configuration of Exemplary Embodiments

(72) The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements can be reversed or otherwise varied and the nature or number of discrete elements or positions can be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps can be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions can be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present

disclosure.

(73) The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure can be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions. Inputs and outputs can be differential, single ended, paired, etc.

(74) Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps can be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

Claims

1. A communication switch comprising: a first magnetic circuit having a first movable core, a first transmit output coupled with a first connector, a first transmit input, and second transmit input; a second magnetic circuit having a second movable core, a first receive input coupled with the first connector, a first receive output, and second receive output; a third magnetic circuit having a second transmit output coupled with a second connector, a third transmit input, and fourth transmit input, the fourth transmit input coupled with the second transmit input, the first transmit input and the third transmit input coupled with a controller; and a fourth magnetic circuit having a second receive input coupled with the second connector, a third receive output, and a fourth receive output, the fourth receive output coupled with the second receive output, the first receive output and the third receive output coupled with the controller; wherein the first transmit output is coupled to the first transmit input by the first movable core and the first receive output is coupled to the first receive input by the second movable core in a first condition, and wherein the first transmit output is coupled to the second transmit input by the first movable core and the first receive input is coupled to the second receive output by the second movable core in a second condition.
2. The communication switch of claim 1, wherein the second condition is power loss in the controller.
3. The communication switch of claim 2, wherein the first transmit output and the first receive input are coupled to a first Ethernet jack.
4. The communication switch of claim 3, wherein the second transmit output and the second receive input are coupled to a second Ethernet jack.
5. The communication switch of claim 3, wherein the third magnetic circuit comprises a third movable core, and the fourth magnetic circuit comprises a fourth movable core; wherein the second

transmit output is coupled to the third transmit input by the third movable core and the third receive output is coupled to the second receive input by the fourth movable core in the first condition, and wherein the second transmit output is coupled to the fourth transmit input by the third movable core and the second receive input is coupled to the fourth receive output by the fourth movable core in the second condition.

6. The communication switch of claim 5, wherein the first magnetic circuit comprises a transformer including the first movable core, the second magnetic circuit comprises a second transformer including the second movable core, the third magnetic circuit comprises a third transformer including the third movable core, and the fourth magnetic circuit comprises a fourth transformer including the fourth movable core.

7. The communication switch of claim 3, wherein the communication switch is a three port Ethernet switch.

8. A communication switch for use in a building, the communication switch comprising: a first circuit having a first transmit output coupled with a first connector, a first transmit input, and second transmit input; a second circuit having a first receive input coupled with the first connector, a first receive output, and second receive output; a third circuit having a second transmit output coupled with a second connector, a third transmit input, and fourth transmit input, the fourth transmit input coupled with the second transmit input, the first transmit input and the third transmit input coupled with a controller; and a fourth circuit having a second receive input coupled with the second connector, a third receive output, and a fourth receive output, the fourth receive output coupled with the second receive output, the first receive output and the third receive output coupled with the controller; wherein the first transmit output is coupled to the first transmit input by a first movable core and the first receive output is coupled to the first receive input by a second movable core in a first condition, and wherein the first transmit output is coupled to the second transmit input by the first movable core and the first receive input is coupled to the second receive output by the second movable core in a second condition.

9. The communication switch of claim 8, wherein the second condition is power loss in the controller.

10. The communication switch of claim 8, wherein the first transmit output and the first receive input are coupled to a first Ethernet jack.

11. The communication switch of claim 8, wherein the second transmit output and the second receive input are coupled to a second Ethernet jack.

12. The communication switch of claim 8, wherein the second transmit output is coupled to the third transmit input by a third movable core and the third receive output is coupled to the second receive input by a fourth movable core in the first condition, and wherein the second transmit output is coupled to the fourth transmit input by the third movable core and the second receive input is coupled to the fourth receive output by the fourth movable core.

13. The communication switch of claim 12, wherein the first circuit comprises a first transformer including the first movable core, the second circuit comprises a second transformer including the second movable core, the third circuit comprises a third transformer including the third movable core, and the fourth circuit comprises a fourth transformer including the fourth movable core.

14. The communication switch of claim 8, wherein the communication switch is a three port Ethernet switch.

15. A communication switch for use in an Ethernet network, the communication switch comprising: a first circuit having a first core, a first transmit output coupled with a first connector, a first transmit input, and second transmit input; a second circuit having a second core, a first receive input coupled with the first connector, a first receive output, and second receive output; a third circuit having a second transmit output coupled with a second connector, a third transmit input, and fourth transmit input, the fourth transmit input coupled with the second transmit input, the first transmit input and the third transmit input coupled with a controller; and a fourth circuit having a

second receive input coupled with the second connector, a third receive output, and a fourth receive output, the fourth receive output coupled with the second receive output, the first receive output and the third receive output coupled with the controller; wherein the first transmit output is coupled to the first transmit input and the first receive output is coupled to the first receive input in a first condition by an action of a control circuit, and wherein the first transmit output is coupled to the second transmit input and the first receive input is coupled to the second receive output in a second condition by the action of the control circuit, wherein the action comprises energizing or deenergizing a coil to move the first core and the second core.

16. The communication switch of claim 15, wherein the second condition is power loss in the controller.

17. The communication switch of claim 15, wherein the first transmit output and the first receive input are coupled to a first Ethernet jack.

18. The communication switch of claim 15, wherein the second transmit output and the second receive input are coupled to a second Ethernet jack.

19. The communication switch of claim 15, wherein the third circuit comprises a third core, and the fourth circuit comprises a fourth core; wherein the second transmit output is coupled to the third transmit input and the third receive output is coupled to the second receive input in the first condition by the action of the control circuit, and wherein the second transmit output is coupled to the fourth transmit input and the second receive input is coupled to the fourth receive output in the second condition by the action of the control circuit.

20. The communication switch of claim 19, wherein the first circuit, the second circuit, the third circuit, and the fourth circuit each comprise a transformer including one of the first core, the second core, or the third core.
